

Econ 646 Spring 2011
Midterm (March 21st)

1. Consider the following problem:

$$\max E_0 \left[\sum_{t=0}^{\infty} \beta^t (\log C_t + \theta \log (1 - N_t)) \right]$$

subject to

$$Y_t + (1 - \delta) K_{t-1} = C_t + K_t$$

$$Y_t = A_t K_{t-1}^{\alpha} (X_t N_t)^{1-\alpha}$$

$$X_t = \gamma^t X_0$$

$$\log A_t = \rho \log A_{t-1} + \varepsilon_t + \eta_{t-1}$$

$$K_{-1} \text{ and } A_0 \text{ are given}$$

ε_t and η_t are white noise. Be careful for timing convention: as in the lecture, subscript of a variable denotes the timing at which the variable is known.

- (a) Derive the first order conditions. Give a brief intuition (a sentence or two) for each equation.
- (b) Write down the system of equations that summarizes this economy. Be careful that the number of endogenous variables must be matched with the number of equations.
- (c) Show that this economy accommodates the balanced growth path. That is, show that if you appropriately define detrended variables, the resulting system does not have X_t any more.
- (d) In (c), what will happen to N_t ? Answer it, and give intuition for this result.
- (e) Assume $\gamma = 1$. Log-linearize the detrended system around the steady state. You have to express all the coefficients with parameters (α, β, δ , etc.) and the steady state labor hours N .
- (f) Suppose in period 0, the economy is at the steady state. Figure 1 plots impulse response functions to a positive realization of $\varepsilon_1 > 0$ and Figure 2 plots impulse response functions to a positive realization of $\eta_1 > 0$, under typical parameter values. In both figures, the vertical axis measures percentage deviation from the steady state. Notice that sharp differences are observed on impact: in Figure 1, everything moves in the same direction, while in Figure 2, consumption rises but labor hours, investment, and output decrease on impact. Explain (intuitively and concisely) why such differences are observed.

